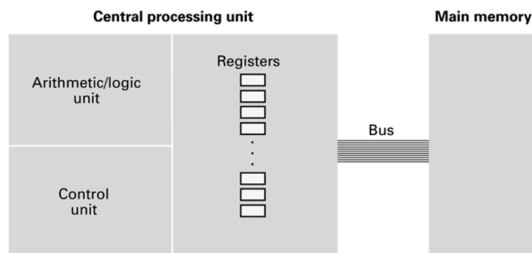


Chapter2 Data Manipulation

Computer Architecture

- **Central Processing Unit(CPU)**
 - Arithmetic/Logic unit & Control unit (ALU & CU)
 - Registers
 1. General Purpose
 2. Special Purpose e.g. Programmer Counter(PC), Instruction Register(IR)
- **Bus**
- **Motherboard**



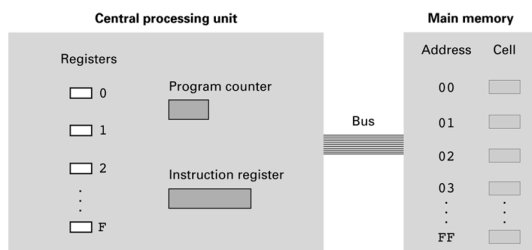
Machine Language Philosophies

- **RISC:** few, simple, efficient, fast e.g. PowerPC from Apple/IBM/Motora and ARM
- **CISC:** many, convenient, power e.g. Intel

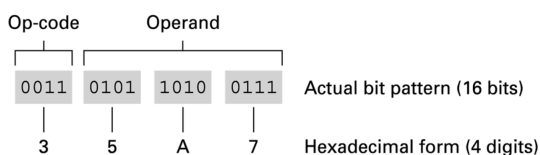
Machine Instruction Types

- **Data Transfer:** LOAD, MOVE...
- **Arithmetic / Logic:** ADD, AND, OR, XOR
- **Control:** JUMP...

Appendix C

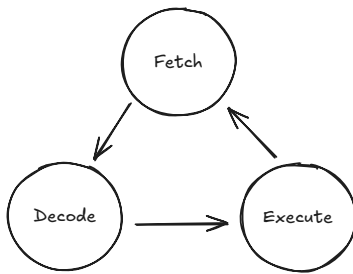


- **The composition of an instruction**

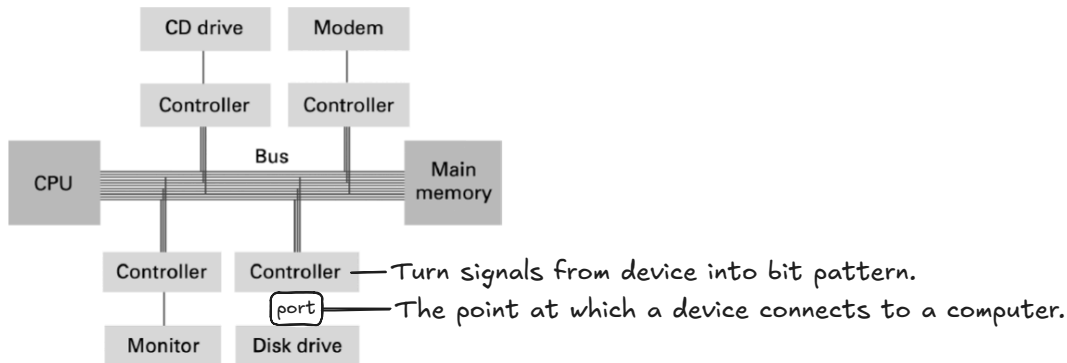


Program Execution

- **Registers(Special-purpose)**
 - Program Counter(PC): Address of next instruction
 - Instruction Register(IR): Current instruction
- **Machine Cycle**
 1. Fetch: CPU requests main memory give it contents of two memory cells. ^[1]
 2. Decode: CPU place the instruction in IR and IR decodes it. In the same time, PC+2. ^[2]
 3. Execute: e.g. LOAD: CPU sends specific signals to Main Memory.; ADD: CPU activates ALU.



Communicating with other device



- **Memory-mapped I/O:** CPU communicates with peripheral devices as though they were memory cells.



- **I/O Instructions:** Special CPU commands used to communicate with I/O devices.
- **Direct memory access (DMA):** Main memory access by a controller over the bus.
 - **Adv:** CPU can do other work in the same time.
 - **Dis-adv:** Von neumann Bottleneck^[3] → let sufficient low
- **Handshaking:** Let CPU know this device's condition
 - Status word: A bit map in which the bits reflect the condition of device.
- **Parallel Communication:** Several communication paths transfer bits simultaneously.
- **Serial Communication:** Bits are transferred one after the other over a single communication path.

Data Communication Rates

- **Measure Units**
 - Bps: bits per second
 - Bandwidth: Maximum available rate

New Forms of Computer Architecture

- **Pipelining:** The steps in the machine cycle to overlap.
 - It means that **many machine cycles can run at the same time**, each at a different stage.
- **Parallel Processing:** Use multiple processors(like CPU) simultaneously.
 - MMID^[4]: Different program, different data^[5]
 - SISD^[6]: Same program, same data → No parallel processing
 - SIMD^[7]: Same program, different data^[8]

1. a instruction is two bytes long ↩
2. next instruction's address ↩
3. CPU and controller can't both use bus in the same time. If CPU is using bus now, controller must wait until CPU finished. ↩
4. Multiple-Instruction Stream, Multiple-Data Stream ↩
5. Multiple processor use same main memory. ↩
6. Single-Instruction Stream, Single-Data Stream ↩

7. Single-Instruction Stream, Multiple-Data Stream ↩

8. Processors are linked to execute the **same instruction sequence simultaneously**. ↩